

Program Evaluation with Confounding Treatment

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Potential Outcomes

Imbens, G. W., & Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica: journal of the Econometric Society*, 467-475.

- Outcome of interest: Y_i
- Binary treatment variable: $D_i \in \{0, 1\}$
- Potential outcome ($Y_i(0), Y_i(1)$):

$$\text{potential outcome} = \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases}$$

- Observed outcome:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

Average Treatment Effect on the Treated: Identification Problem

- The observed difference in outcomes:

$$\begin{aligned}\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \underbrace{\mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 1]}_{\text{ATT}} \\ &\quad + \underbrace{\mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]}_{\text{selection bias ("confounding treatment")}}\end{aligned}$$

- The term $\mathbb{E}[Y_i(0)|D_i = 1]$ captures the **counterfactual** potential outcome of the untreated group if they were treated
- Without further assumptions, the ATT is not identified.
- Randomized control trial (RCT): $Y_i(0)$ and D_i are independent so that selection bias is eliminated

Regression Analysis under Randomization

We can write the outcome variable as an equation of the treatment dummy D_i :

$$Y_i = \underbrace{\alpha}_{=\mathbb{E}(Y_i(0))} + \underbrace{\rho}_{=Y_i(1)-Y_i(0)} D_i + \eta_i$$

so that

$$\begin{aligned}\mathbb{E}[Y_i|D_i = 1] &= \alpha + \rho + \mathbb{E}[\eta_i|D_i = 1] \\ \mathbb{E}[Y_i|D_i = 0] &= \alpha + \mathbb{E}[\eta_i|D_i = 0] \\ \mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] &= \rho + \underbrace{\mathbb{E}[\eta_i|D_i = 1] - \mathbb{E}[\eta_i|D_i = 0]}_{\text{selection bias}}\end{aligned}$$

Under randomization, selection bias should be zero. Otherwise, D_i will be correlated with η_i (endogeneity).

Instruments

A valid instrument is as good as randomly assigned: it is independent of the *potential outcomes* and *potential treatment assignments*.

- For simplicity, suppose that the instrument $Z_i \in \{0, 1\}$ is binary
- $D_i(0)$ and $D_i(1)$ are the *potential treatment assignments* when Z_i takes values of 0 and 1, respectively
- **Condition 1** in the paper:
 - ① $(Y_i(0), Y_i(1), D_i(0), D_i(1))$ are jointly independent of Z_i
 - ② $P(w) = \mathbb{E}[D_i|Z_i = w]$ is a nontrivial function of w , i.e. $\mathbb{E}[D_i(1) - D_i(0)] \neq 0$

Deriving Local Average Treatment Effect

$$\begin{aligned} & \mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0] \\ &= \mathbb{E}[D_i(1)Y_i(1) + (1 - D_i(1))Y_i(0)|Z_i = 1] - \mathbb{E}[D_i(0)Y_i(1) + (1 - D_i(0))Y_i(0)|Z_i = 0] \\ &= \mathbb{E}[(D_i(1) - D_i(0))(Y_i(1) - Y_i(0))] \tag{1} \\ &= \Pr(D_i(1) - D_i(0) = 1)\mathbb{E}[Y_i(1) - Y_i(0)|D_i(1) - D_i(0) = 1] \tag{2} \\ &\quad - \Pr(D_i(1) - D_i(0) = -1)\mathbb{E}[Y_i(1) - Y_i(0)|D_i(1) - D_i(0) = -1] \tag{3} \end{aligned}$$

Remarks

- (1) is derived by condition 1
- Without further restrictions, cannot separately identify (2) from (3)
- LHS is observed, but on the RHS, there are two unknowns

Condition 2: Monotonicity

Imbens and Angrist (1994) introduces the following **monotonicity assumption** to overcome the identification problem:

- In this case, either $D_i(1) \geq D_i(0)$ for all i , or $D_i(1) \leq D_i(0)$ for all i
so that either line (2) or line (3) is eliminated
- Remark: Intuitively this is a “no defiers” assumption, but it is empirically untestable

Local Average Treatment Effect

Under conditions 1 and 2, the LATE is identified. Suppose $D_i(1) \geq D_i(0)$ for all i , the LATE is

$$\mathbb{E}[Y_i(1) - Y_i(0) | D_i(1) > D_i(0)] = \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{P(1) - P(0)} = \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0]}$$

- The LATE is the treatment effect of a subgroup of people whose treatment statuses are changed by the instrument (“Compliers”)
- But we do NOT know who belongs to this subgroup
- This is the probability limit of the IV estimator (Regress Y on D , with Z as an IV for D).

General Case

Let $\mathcal{Z}$ be the support of Z .

- **Condition 1 (Existence of Instruments):** For all $w \in \mathcal{Z}$,
 - ① the triple $(Y_i(0), Y_i(1), D_i(w))$ is jointly independent of Z_i .
 - ② $P(w) = E[D_i | Z_i = w]$ is a nontrivial function of w .
- **Condition 2 (Monotonicity):** For all $z, w \in \mathcal{Z}$, either $D_i(z) \geq D_i(w)$ for all i , or $D_i(z) \leq D_i(w)$ for all i .

Theorem

If **Conditions 1 and 2** hold, then we can identify the following *local average treatment effect*:

$$\alpha_{z,w} = E[Y_i(1) - Y_i(0) | D_i(z) \neq D_i(w)]$$

from the joint distribution of Y , D , and Z , for all z and w in $\mathcal{Z}$ such that $E[Y_i | Z_i = z]$ and $E[Y_i | Z_i = w]$ are finite, and $P(z) \neq P(w)$.

General Case

• **Condition 3:** $g(z)$ is a function from $\mathcal{Z}$ to $\mathfrak{R}$ such that

- 1 either for all $z, w \in \mathcal{Z}$, $\Pr(z) \leq P(w)$ implies $g(z) \leq g(w)$, or, for all $z, w \in \mathcal{Z}$, $P(z) \leq P(w)$ implies $g(z) \geq g(w)$;
- 2 $\text{Cov}(D, g(Z)) \neq 0$.

Theorem

Suppose that **Conditions 1, 2, and 3** are satisfied. Let $\mathcal{Z} = \{z_0, z_1, \dots, z_K\}$, ordered in such a way that if $l < m$ then $P(z_l) \leq P(z_m)$. Then, the IV estimator estimates

$$\alpha_g^{IV} = \text{Cov}(Y, g(Z)) / \text{Cov}(D, g(Z)) = \sum_{k=1}^K \lambda_k \cdot \alpha_{z_k, z_{k-1}},$$

with weights

$$\lambda_k = \frac{(P(z_k) - P(z_{k-1})) \cdot \sum_{l=k}^K P(Z = z_l) \cdot (g(z_l) - E[g(Z)])}{\sum_{m=1}^K (P(z_m) - P(z_{m-1})) \cdot \sum_{l=m}^K P(Z = z_l) \cdot (g(z_l) - E[g(Z)])}$$

Further Issues and Examples

- Angrist (1990) estimates the causal effect of veteran status on earnings, using as instruments the draft lottery outcomes
- Instrument strength in IV estimation and inference:
 - Keane, M., & Neal, T. (2023). Instrument strength in IV estimation and inference: A guide to theory and practice. *Journal of Econometrics*, 235(2), 1625–1653.
 - Keane, M. P., & Neal, T. (2024). A Practical Guide to Weak Instruments. *Annual Review of Economics*, 16, 185–212.